

Estimating the Height of a Glider from its Shadow in an Aerial Orthophoto

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Abstract

A glider and its shadow both appear in a SWISSIMAGE orthophoto tile near Isenau, Switzerland. This note sets out the geometry and mathematics used to turn that observation into a height above ground. The method needs three inputs: the horizontal separation between the glider and its shadow, the terrain elevations beneath each, and the solar elevation angle. The solar angle is obtained in two independent ways: by calibration against vertical reference masts of estimated height, and from a solar ephemeris evaluated at the documented flight date. Uncertainty is carried through with worst-case interval arithmetic, which is exact for the monotone formulas involved. We also quantify the main error that the intervals do not cover: orthophoto displacement of an elevated object. Taking the ephemeris route and the glider's apparent position, the central estimate is about 142 m above the terrain beneath the glider. This figure is conditional on assumptions stated explicitly below, and the uncorrected orthophoto displacement could shift it by tens of metres.

1 Problem and approach

An orthophoto is an aerial image resampled onto a terrain model, so that every ground point appears at its map position. An object in the air, such as a glider, breaks this assumption twice. It casts a shadow some distance away on the ground, and the object itself is drawn at a position that is displaced by the projection. The shadow gives the height, and the displacement is the main thing that can spoil it.

The calculation has four steps:

1. Measure, in a common horizontal frame, the glider position and the matching point of its shadow (Section 3).
2. Read the terrain elevation at both points (Section 2).
3. Determine the solar elevation angle α , either from reference masts (Section 4) or from a solar ephemeris (Section 5).
4. Propagate the input bounds to a height interval (Section 6) and assess what those bounds leave out (Section 7).

The browser calculator in this repository implements steps 1–4 with mast calibration or a manually entered solar angle. The ephemeris numbers are reproduced by `paper/solar.mjs`.

2 Data and reference frames

Horizontal coordinates are in CH1903+/LV95 (EPSG:2056), a conformal projection. Combining its grid scale with the reduction from about 1.9 km elevation to the ellipsoid gives a factor within about 3×10^{-4} of unity at Isenau. LV95 coordinate differences can therefore be treated as horizontal ground distances, with an error under 3 cm over the 76 m baseline used here. Elevations come from the swisstopo terrain-height service (swissALTI3D). Only differences between nearby elevations enter the formulas, so the vertical datum cancels as long as one is used throughout.

The terrain model is bare-earth. Where vegetation, buildings or snow raise the surface that actually catches a shadow, the effective shadow elevation is higher than the model value. Section 6 treats this as part of the terrain bound ε_z .

3 Shadow geometry

3.1 Parallel-ray model

At the scale involved (less than 1 km), sunlight is modelled as a family of parallel straight rays with elevation α above the horizontal and azimuth $A_\odot$. Three effects are neglected:

- The Sun's 0.53° disk makes the shadow edge soft without moving its centre; see Section 7.
- Earth curvature contributes $D^2/2R \approx 0.5$ mm for $D = 76$ m and $R = 6\,371$ km.
- Atmospheric refraction is below $30''$ at $\alpha \approx 65^\circ$.

Let $P = (E_g, N_g, Z)$ be a point on the glider and $S = (E_s, N_s, z_s)$ the corresponding point of its shadow on the terrain. The ray through P reaches the ground at S . Its horizontal projection runs from (E_g, N_g) to (E_s, N_s) , and its vertical drop is $Z - z_s$, so

$$\tan \alpha = \frac{Z - z_s}{D}, \quad D = \sqrt{(E_s - E_g)^2 + (N_s - N_g)^2}. \quad (1)$$

The shadow lies on the side away from the Sun, so its azimuth from the glider is

$$A_{gs} = \text{atan2}(E_s - E_g, N_s - N_g) \equiv A_\odot + 180^\circ \pmod{360^\circ}. \quad (2)$$

Equation (2) does not enter the height computation, but it is a useful independent consistency check (Section 8).

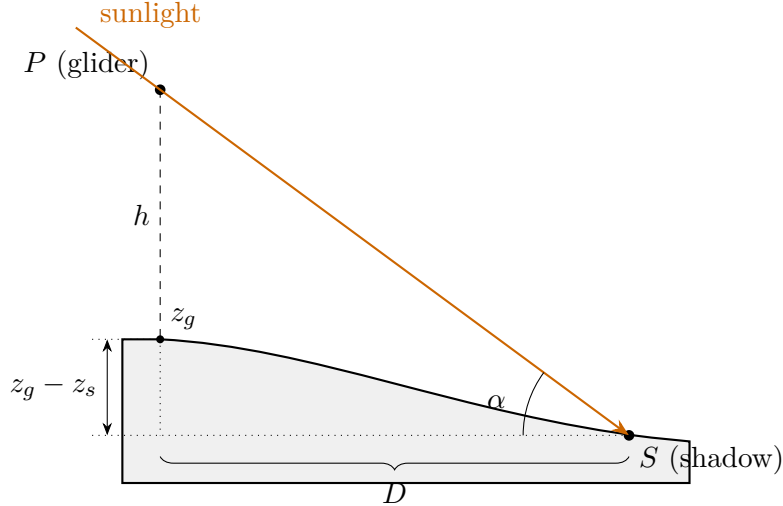


Figure 1: Vertical section along the sunlight direction. On sloping terrain the shadow falls lower than the ground beneath the glider ($z_s < z_g$). The terrain difference therefore has to be removed from the height obtained along the ray.

3.2 Height above ground

The height above the terrain directly beneath the glider is $h = Z - z_g$, where z_g is the terrain elevation at (E_g, N_g) . Eliminating Z with (1) gives the central formula

$$h = D \tan \alpha + z_s - z_g \quad (3)$$

The term $z_s - z_g$ is the terrain correction. A shadow that falls downhill ($z_s < z_g$) would overstate the height if the correction were ignored, and one that falls uphill would understate it. On flat ground (3) reduces to $h = D \tan \alpha$.

The height is referred to the terrain directly beneath the glider, which is what *above ground level* means. If the height is wanted above some other reference, such as the shadow point or sea level, replace z_g in (3) with that reference.

4 Solar elevation from reference masts

4.1 Calibration equation

A vertical mast of height H , with base elevation z_b , casts its top's shadow at horizontal distance L from the base, onto terrain at elevation z_t . Applying (1) to the mast top gives

$$\tan \alpha = \frac{H + z_b - z_t}{L}. \quad (4)$$

Substituting (4) into (3) gives a combined estimator,

$$h = \frac{D}{L} (H + z_b - z_t) + z_s - z_g. \quad (5)$$

In (5) the horizontal lengths enter only as the ratio D/L . A scale error common to both, such as a wrong image resolution, therefore cancels. This holds only if the mast and glider were imaged under the same sun. Rays are parallel only at a single instant, so the mast shadow and the glider shadow must come from the same exposure, or at least from exposures taken within minutes of each other. Tiles of the same mosaic year do not establish this.

The mast top must be a point vertically above the base used for z_b . A point on an offset crossarm needs its own horizontal projection. A second mast gives an independent value of α . The calculator compares it with the first value and does not average the two. Averaging would hide a disagreement that may point to a wrong shadow identification or to different capture times.

4.2 Mast heights from ground photographs

No surveyed mast heights were available. Each height was reconstructed from a March 2011 ground panorama by elevation-angle triangulation. The camera stands at terrain elevation z_c with lens height ℓ above it and horizontal distance d from the mast. If the mast top is seen at angle β above the horizontal,

$$H = z_c + \ell + d \tan \beta - z_b. \quad (6)$$

4.3 How mast errors propagate

From (5), $\partial h / \partial H = D/L = \tan \alpha \cdot D/(H + z_b - z_t)$. Near $\alpha \approx 65^\circ$ a 9 m mast on level ground casts a shadow of only $L \approx 4.2$ m. The amplification factor is then $D/L \approx 18$, so a ± 3 m uncertainty in H spreads the glider height over roughly 88 m to 197 m. That spread comes before any error in L , which at 10 cm ground resolution is itself a few percent. Short shadows under a high summer sun make mast calibration intrinsically weak for this scene. This motivates the ephemeris route described next.

Input	Mast 1	Mast 2
Terrain at camera z_c	1 790.3 m	1 776.1 m
Assumed lens height ℓ	2.5 m	2.5 m
Horizontal distance d	24.75 m	40.84 m
Top elevation angle β	7.8°	17.4°
Terrain at mast base z_b	1 787.2 m	1 781.4 m
H from (6)	8.99 m	10.00 m
$\partial H / \partial \beta = d \sec^2 \beta$	0.44 m/°	0.78 m/°
Working bound	± 3 m	± 3 m

Table 1: Street-View reconstruction of the mast heights. The lens height is an assumption: $\partial H / \partial \ell = 1$, so it passes one-for-one into H . The ± 3 m figure is a working sensitivity bound, not a measured tolerance.

5 Solar elevation from an ephemeris

5.1 Acquisition time

The covering SWISSIMAGE flight strip is 20230714_1118_12504, flown on 14 July 2023. Its identifier encodes the *start* of acquisition, 11:18 UTC. The exposure at the glider comes later along the strip and is not published, so the solar angle has to be bounded over a time window rather than evaluated at one instant. The strip’s whole extent is about 58 km east–west. At typical survey speeds that takes on the order of a quarter of an hour, and we adopt the generous window 11:18–11:48 UTC.

5.2 Solar position

The solar position follows the low-precision algorithm of Meeus, as used in the NOAA solar calculator, which is accurate to about $1'$ for present-day dates. With T the number of Julian centuries since J2000.0, the steps are as follows:

$$L_0 = 280.466\,46^\circ + 36\,000.769\,83^\circ T + 0.000\,303\,2^\circ T^2 \quad (\text{mean longitude}) \quad (7)$$

$$M = 357.529\,11^\circ + 35\,999.050\,29^\circ T - 0.000\,153\,7^\circ T^2 \quad (\text{mean anomaly}) \quad (8)$$

$$\lambda = L_0 + C(M) - 0.005\,69^\circ - 0.004\,78^\circ \sin \Omega \quad (\text{apparent longitude}) \quad (9)$$

$$\sin \delta = \sin \varepsilon \sin \lambda \quad (\text{declination}) \quad (10)$$

Here C is the equation of centre, Ω the longitude of the Moon’s ascending node and ε the obliquity of the ecliptic. The local hour angle $H_\odot$ follows from UTC, the equation of time and the longitude λ_{geo} . With latitude φ , elevation and azimuth are

$$\sin \alpha = \sin \varphi \sin \delta + \cos \varphi \cos \delta \cos H_\odot, \quad (11)$$

$$A_\odot = \text{atan2}(\sin H_\odot, \cos H_\odot \sin \varphi - \tan \delta \cos \varphi) + 180^\circ. \quad (12)$$

At the glider ($\varphi = 46.3674^\circ$ N, $\lambda_{\text{geo}} = 7.2103^\circ$ E, converted from LV95 with the swisstopo approximation and cross-checked against the swisstopo REFRAME service), the declination is $\delta = 21.68^\circ$, the equation of time is -5.9 min, and local solar noon falls at 11:37 UTC.

Near solar noon $d\alpha/dt \approx 0$, so a modest error in the acquisition time hardly affects the elevation. Over the whole window 11:18–11:48 UTC, $\alpha \in [65.01^\circ, 65.31^\circ]$, i.e. $\alpha = 65.16^\circ \pm 0.15^\circ$. In contrast, the azimuth turns at about 0.5° per minute, which makes it a sensitive clock (Section 8).

UTC (2023-07-14)	α	$A_\odot$	$\tan \alpha$
11:00	64.19°	159.92°	2.0676
11:18 (strip start)	65.01°	169.49°	2.1452
11:24	65.16°	172.74°	2.1600
11:37 (solar noon)	65.31°	179.98°	2.1748
11:48	65.21°	186.08°	2.1647
12:18	63.94°	202.09°	2.0446

Table 2: Solar position at the glider. An independent evaluation with ASTROPY agrees to within 0.01° in both angles.

6 Bounded uncertainty propagation

6.1 Interval arithmetic for monotone formulas

Each input x is given as a central value with a non-negative bound, $x \in [x - \varepsilon_x, x + \varepsilon_x]$. The calculator reports the exact range of the output over the box of all inputs. This is not a statistical confidence interval.

Proposition 1. *Let $f(x_1, \dots, x_n)$ be monotone in each argument over a box $B = \prod_i [a_i, b_i]$. Then $\min_B f$ and $\max_B f$ are attained at the corner where each x_i is set to the end of its interval that minimises or maximises f . That end is fixed by the sign of $\partial f / \partial x_i$.*

Proof. Change one coordinate at a time towards the chosen endpoint. By monotonicity no step decreases (or increases) f , so the chosen corner is extremal. $\square$

Height formula. With $D > 0$ and $\tan \alpha > 0$, h in (3) increases with D , $\tan \alpha$ and z_s and decreases with z_g . The terrain elevations z_s and z_g are bounded independently, each by ε_z . Hence

$$h_{\min} = (D - \varepsilon_D) \tan \alpha_{\min} + (z_s - z_g) - 2\varepsilon_z, \quad (13)$$

$$h_{\max} = (D + \varepsilon_D) \tan \alpha_{\max} + (z_s - z_g) + 2\varepsilon_z, \quad (14)$$

provided $D - \varepsilon_D > 0$. Because $\tan$ is increasing on $(0^\circ, 90^\circ)$, a direct solar bound $\alpha \pm \varepsilon_\alpha$ maps to $[\tan(\alpha - \varepsilon_\alpha), \tan(\alpha + \varepsilon_\alpha)]$. The calculator requires $0 < \alpha - \varepsilon_\alpha$ and $\alpha + \varepsilon_\alpha < 90^\circ$.

Mast calibration. Write $R = H + z_b - z_t$ for the rise. Then $\tan \alpha = R/L$ increases in R and decreases in L , and $\varepsilon_R = \varepsilon_H + 2\varepsilon_z$, so

$$\tan \alpha_{\min} = \frac{R - \varepsilon_R}{L + \varepsilon_L}, \quad \tan \alpha_{\max} = \frac{R + \varepsilon_R}{L - \varepsilon_L}, \quad (15)$$

which require $R - \varepsilon_R > 0$ and $L - \varepsilon_L > 0$. These conditions mean the Sun must stay above the horizon and the shadow must have positive length throughout the box. Chaining the two monotone maps gives the overall height interval.

The worst case treats z_s , z_g (and z_b , z_t) as independent. If their errors are correlated, for example through a common datum offset, the true spread is smaller, so the interval is conservative. Bounds set to zero mean that the uncertainty is *not quantified*, not that the input is exact.

6.2 First-order sensitivities

Linearising (3) shows which input dominates:

$$\frac{\partial h}{\partial D} = \tan \alpha, \quad \frac{\partial h}{\partial \alpha} = D \sec^2 \alpha, \quad \frac{\partial h}{\partial z_s} = 1, \quad \frac{\partial h}{\partial z_g} = -1. \quad (16)$$

At $\alpha = 65.16^\circ$, $\tan \alpha = 2.16$ and $\sec^2 \alpha = 5.66$. For $D = 76$ m:

- 1 m of horizontal error in D moves h by 2.16 m;
- 1° of solar error moves h by $D \sec^2 \alpha \cdot \pi/180 = 7.5$ m;
- each metre of terrain error moves h by 1 m.

A high Sun amplifies errors in the measured separation. Accuracy therefore depends on locating the glider and its shadow correctly far more than on the exact timestamp.

7 Error sources outside the bounds

7.1 Orthophoto displacement of the glider

The orthophoto places every pixel as if it lay on the terrain. A point at height h above the terrain, viewed at zenith angle θ from the camera, is therefore drawn displaced *away* from the camera's nadir by

$$\Delta = h \tan \theta. \quad (17)$$

The shadow lies on the terrain and is not displaced. Only the component of Δ along the shadow direction changes D . Let $\theta_{\parallel}$ be the signed view tilt projected onto the vertical plane of the sunlight, positive when the glider is drawn towards its shadow. The observed separation is then $D_{\text{obs}} = D - h \tan \theta_{\parallel}$. Substituting $D = D_{\text{obs}} + h \tan \theta_{\parallel}$ into (3) and solving for h gives the self-consistent correction

$$h = \frac{D_{\text{obs}} \tan \alpha + z_s - z_g}{1 - \tan \theta_{\parallel} \tan \alpha}. \quad (18)$$

The amplification is strong. With $\tan \alpha \approx 2.16$, a view tilt of only $\pm 5^\circ$ along the ray changes h by roughly -16% to 23% . That is about -23 m to 33 m at $h \approx 142$ m. The camera pose of the source exposure is not reconstructed here, so $\theta_{\parallel}$ is unknown. Equation (18) is the place to apply it once it is known. The calculator uses the apparent glider position and flags its results as provisional for this reason. The terrain value z_g should also be re-read beneath the corrected glider position.

The displacement component *perpendicular* to the shadow ray would show up as a mismatch between the observed azimuth A_{gs} and $A_{\odot} + 180^\circ$ at the exposure time. Section 8 finds no such mismatch within the time window. The along-ray component, which is the one that affects h , cannot be detected that way.

7.2 Other effects

- **Finite solar disk.** The Sun subtends ≈ 9.3 mrad, so a shadow cast from slant range $h/\sin \alpha$ is blurred by about 1.6 m along the ground. This softens the shadow edges but leaves the centroid unbiased. Matching corresponding points (e.g. wing centre to shadow centre) avoids edge bias.
- **Shadow identification.** If the shadow belongs to a different object or a different exposure, the result is meaningless. No interval covers this.
- **Mosaic time mixing.** A mosaic may combine exposures taken at different times. Mast calibration is valid only if masts and glider share one exposure.
- **Terrain epoch and surface.** The terrain service reflects the present bare-earth model, not necessarily the surface at acquisition (snow, vegetation, works).

8 Case study: Isenau, 14 July 2023

Geometry. From Table 3, $D = 75.959$ m, $A_{gs} = 352.74^\circ$ and $z_s - z_g = -21.4$ m. The shadow falls about 21 m downhill, so the terrain correction lowers the height.

Point	Easting	Northing	Terrain
Glider (apparent)	2 582 425.46	1 135 134.94	1 927.8 m
Shadow (candidate)	2 582 415.86	1 135 210.29	1 906.4 m

Table 3: LV95 coordinates of the map pins and terrain elevations (retrieved 7 October 2026). These are pin centres, not surveyed points.

Azimuth consistency. Equation (2) requires $A_{\odot} = 172.74^{\circ}$, which the ephemeris gives at 11:24 UTC. That is six minutes after the strip start and inside the adopted window. The shadow direction is therefore consistent with the documented flight, and there is no evidence of a cross-ray displacement. A shadow cast an hour earlier or later would point about 30° away.

Height (ephemeris route). With $\alpha = 65.16^{\circ} \pm 0.15^{\circ}$ and the apparent glider position,

$$h = 75.959 \times \tan 65.16^{\circ} - 21.4 \approx 142.7 \text{ m}, \quad h \in [141.5 \text{ m}, 143.8 \text{ m}] \text{ (solar bound only)}. \quad (19)$$

Adding plausible measurement bounds of $\varepsilon_D = 5 \text{ m}$ and $\varepsilon_z = 1 \text{ m}$ widens this to 128.8 m to 156.7 m. Even a $\pm 1^{\circ}$ solar bound, wide enough to cover acquisition anywhere from about 11:00 to 12:15 UTC, gives 135.4 m to 150.5 m with exact D and terrain.

Height (mast route). The aerial shadow lengths of the two masts have not yet been measured. The mast route therefore yields no number, and with $\pm 3 \text{ m}$ mast heights it would be much weaker than the ephemeris (Section 4). Measured mast shadows remain valuable as an independent check that the solar angle in the actual exposure is about 65° .

Summary. Taking the apparent glider position, the candidate shadow and the documented flight date, the glider was about 140 m above the terrain beneath it. The result is conditional on:

- (i) correct shadow identification;
- (ii) exposure during the 2023-07-14 strip near midday;
- (iii) a negligible along-ray orthophoto displacement.

Condition (iii) is the one that limits the accuracy. Equation (18) shows how to correct for it once the source exposure geometry is known.

9 Implementation and verification

Equation	Implementation
(1), (2)	<code>separation()</code>
(4) + bounds	<code>calibrateReference()</code>
direct $\alpha \pm \varepsilon_{\alpha}$	<code>directSun()</code>
(3) + bounds	<code>heightFromShadow()</code>
whole pipeline, warnings	<code>solve()</code>
(11) ff., LV95→WGS84	<code>paper/solar.mjs</code>

Table 4: Where each equation is implemented. Functions without a path are in `glider-height/dist/calculations.mjs`.

The calculation module has no dependencies. Its test suite (`npm test` in `glider-height/`) checks the following:

- the LV95 separation and azimuth of the case-study pins;
- $h = D$ on flat ground at 45° ;
- the sign of the uphill and downhill corrections;
- the corner-case interval bounds against hand-computed values;
- rejection of non-finite, singular and horizon-grazing inputs;
- non-averaging of the second mast;
- flagging, rather than clamping, of below-ground results.

The solar positions in Table 2 were cross-checked against ASTROPY (agreement better than 0.01°). The coordinate conversion was cross-checked against the swisstopo REFRAME service (agreement better than 1 m).

Data sources

- swisstopo SWISSIMAGE orthophoto mosaic, and flight-strip metadata for strip 20230714_1118_12504 from the layer `ch.swisstopo.lubis-bildstreifen`.
- swisstopo swissALTI3D terrain model via the `geo.admin.ch` height service.
- Ground panoramas (March 2011) for the mast-height reconstruction; linked, not reproduced.
- J. Meeus, *Astronomical Algorithms*, 2nd ed., Willmann-Bell, 1998; NOAA Global Monitoring Laboratory solar calculator.